Ship AI Features Without an Inference Bill

Open-weight models on the user's device,
from a web page.

navigator.models

Chris McKenzie · Decentralized AI Day New York · October 10, 2026

Five costs of hosted inference

Spend	The bill grows with usage. Unpredictable spend.
Latency	Every interaction waits for a network round trip.
Availability	An outage or an offline user can disable the feature.
Privacy	The prompt leaves the device, along with the data it contains.
Control	The provider sets prices, limits, terms and retirement dates.

The bill is only part of what your app depends on.

What losing control costs you

OctoAI
October 2024

After Nvidia's acquisition, customers had roughly a month to migrate before the inference service closed.

Windsurf
June 2025

Anthropic cut nearly all direct Claude 3.x capacity with less than a week's notice, according to Windsurf.

OpenAI Pro 200
September 2026

Included Work and Codex usage drops from 20× to 10× Plus. The price stays $200/month.

Disrupted features. Emergency migrations. Trust to rebuild.

These costs shape
what gets built
and who gets to build it

The browser already runs your code

1989

Sir Tim Berners-Lee invented
the World Wide Web at CERN.

Open no-gatekeeper design of the web has turned it into the largest software platform.

JavaScript runs on the user’s computer.
Open models can use the same foundation.

Web Models API, implemented in Common

navigator.models

Open weights	Run the model on the user's device. Download once and share across sites.
Named models	The page requests a model by ID. The user authorizes access for that site.
No account	No API key, provider account or inference subscription between the page and the model.

Common is the Chromium fork that implements this proposed API.

The permission boundary is the point

“example.com may use model X”

Another site asks again.

Another model asks again.

The browser mediates. The user decides.

Live

A support chat that reads this page

About twenty lines. Wi-Fi off.

What you just saw

No server

Inference can work offline after setup.

No API key

No provider credential to leak or revoke.

No account

No provider sign-up before inference.

No inference bill

No provider fee for each conversation.

The page text stayed on the laptop. Control still depends on the design.

navigator.models

```
await navigator.models.requestAccess(id)

await navigator.models.generate(id, { messages })

for await (const c of navigator.models.stream(
  id, { messages }
)) { /* consume chunks */ }

await navigator.models.embed(id, { input })
```

Model ID first

Explicit, scoped access

Caller-owned context

Caller-owned tool loop

The browser handles downloads, storage, runtime selection and memory.

Prompt API and control

Browser	Browser-provided model
Chrome	Gemini Nano
Edge preview	Phi-4-mini

Model choice

The browser supplies the available models. Apps cannot pin arbitrary model IDs.

Access

Chrome documents user activation and iframe policy, rather than a per-site, per-model user grant.

Context

Sessions retain turns and evict older context. Apps can rebuild a curated history.

It makes local AI easier to use, but leaves key choices with the browser.

Two browser engines oppose the Prompt API

Mozilla

Position: negative

Prompts tuned to one model can force other browsers to reproduce its quirks.

A web API should not depend on one company's model-use policy.

WebKit

Position: oppose

Interoperability
Portability
Privacy

These are the concerns tagged on WebKit's position.

A named model makes the dependency explicit and gives developers a choice.

Local inference is a choice per task

Task or condition	Best fit
Sort and prioritize incoming email	On-device
Summarize private notes	On-device
Rewrite a sentence while editing	On-device
Research a complex question across many sources	Cloud
Monitor news and prepare an overnight briefing	Cloud

There are real problems left

Model identity and registries

What does model-id mean?

Usage limits

Who sets the budget and enforces it?

Resource contention

Memory, battery and thermal throttling

Decentralizing the power

Developers choose Pick the open model that fits the task.

Browsers execute Manage hardware, runtime and memory.

Users authorize Control access for each site and model.

The proposal is early

Developers choose.
Browsers execute.
Users authorize.

Review the spec. Challenge its assumptions.
Help define the browser's boundaries.

Thank you



webmodels.dev